
DV2626/2599 Machine Learning

Lecture 1: Introduction

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(some slides from Niklas Lavesson)

Outline

- Presentation of the course
- Presentation of the staff
- Introduction to Machine Learning

Presentation of the course

- DV2626/DV2599 Machine Learning, 7.5/6 ECTS
 - 10 two-hour lectures
 - 2 four-hour exercises (seminar and lab)
 - 3 mandatory assignments (can work in a group of two)
- Main literature: P. Flach (2012). *Machine Learning: The Art and Science of Algorithms that Make Sense of Data*. Cambridge University Press.
- Supplementary literature: A. Burkov (2019). *The Hundred-Page Machine Learning Book*. ISBN 978-1-9995795-0-0, Publisher: Andriy Burkov

List of Lectures

1. Introduction
2. Foundations of Machine Learning
3. Data Preprocessing
4. Logical Models
5. Linear Models and Extended Linear Models
6. Meta and Ensemble Learning
7. Unsupervised Learning and Clustering
8. Artificial Neural Networks
9. Weakly Supervised Learning and Multi-instance Learning
10. Outlier Detection Techniques

List of Exercises

1. Project Idea (seminar, DV2626 will be divided in two groups)
2. ML in Python (lab, divided in two groups)

Coursework

- Assignment 1, 1 ECTS, graded P/F
- Assignment 2, 1 ECTS, graded P/F
- Project, 5.5 ECTS (DV2626) and 4 ECTS (DV2599), graded F-A
 - The grading on the project determines the official grade on the complete course, on the condition that the earlier assignments have been passed

Presentation of the staff

- Course responsible: Prof. Veselka Boeva, veselka.boeva@bth.se
- Lectures
 - L1, L2, L3, L4, L7, L9: Prof. Veselka Boeva
 - L5, L6, L8: Assoc. Prof. Hüseyin Kusetogullari, huseyin.kusetogullari@bth.se
 - L10: Dr. Shahrooz Abghari, shahrooz.abghari@bth.se
- Seminars
 - Veselka Boeva and Shahrooz Abghari
- Labs
 - Shahrooz Abghari

What is Artificial Intelligence (AI)?

John McCarthy - 1950:

“The science and engineering of making **intelligent machines**, especially **intelligent computer programs**”

- AI is a study of how human brain thinks, learns, decides and works, when it tries to solve problems and finally this study outputs **intelligent software systems**.
- The aim of AI is to improve computer functions which are related to human knowledge, e.g., **reasoning**, **learning**, and **problem-solving**.

Traditional software is powered by **code**, whereas AI systems are built using both **code** (**models + algorithms**) and **data**

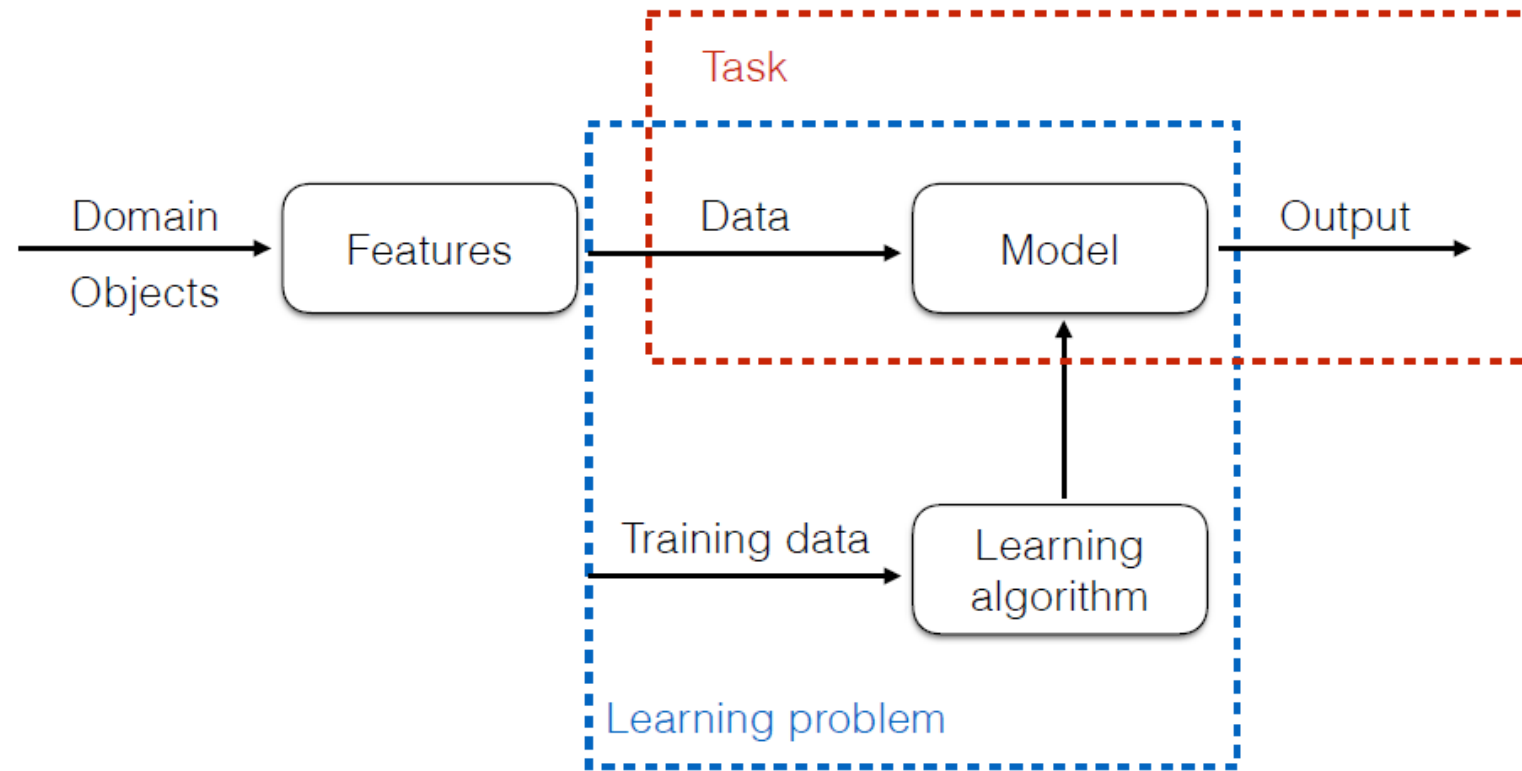
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- Data is **the critical infrastructure** necessary to build AI systems
 - Data largely determines **performance, fairness, robustness, safety, and scalability** of AI systems
 - AI developers understand that **data quality matters**, often spending inordinate amounts of time on data tasks
 - In practice, most organisations fail to create or meet any **data quality standards**
 - The inability of AI projects and ML models to succeed when they are deployed in the real world

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- Andrew Ng (co-founder Coursera & deeplearning.ai) launched a campaign for data-centric AI, 16/06/2021:
 - *When a system isn't performing well, many teams instinctually try to improve the code. But for many practical applications, it's more effective instead to **focus on improving the data**, ...*
 - *The established **model-centric approach to AI** calls for holding the data fixed and iteratively improving the model until the desired results are achieved ...*
 - *In the nascent **data-centric approach to AI** "consistency of data is paramount". To get to the right results, you hold the model or code fixed and iteratively improve the quality of the data*

Introduction to Machine Learning

- What is machine learning?
 - The systematic study of algorithms and systems that improve their knowledge or performance with experience (Flach, 2012)
- Machine learning plays a key role in artificial intelligence, but so do other areas, such as **automated reasoning**
- What is it used for?
 - Anomaly detection, candidate ranking and grouping, data mining, image recognition, information retrieval, natural language processing, object recognition, speech recognition, time series prediction, and many more applications

How machine learning works



A task (red box) requires an appropriate mapping - a model - from data described by features to outputs. Obtaining such a mapping from training data is what constitutes a learning problem (blue box) - Flach (2012)

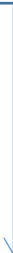
Example task: spam detection (1)

- Suppose we want to address e-mail **spam detection**
- In this domain, the objects are e-mails and the training data are e-mails labeled with a correct output of either *spam* or *ham* (*non-spam*)
- We need to define suitable features to represent e-mails
 - We scan the text body of all the e-mails from the training data and add each word found to a list, which we can call feature names
 - We define each feature as a Boolean, which can be either true (if the corresponding word exists in an e-mail) or false
 - We can now represent the e-mails in the training data as a list of word occurrence vectors, where each e-mail has its own pattern of true and false depending on which words occur in the e-mail in question
- Now, we need to **generate a model that classifies e-mails correctly as *spam* or *ham***

Example task: spam detection (2)

Hello, It's me. Bye /Joe

Original e-mail text body



hello	its	me	bye	joe	i	anna	so	why	an
TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE

Word occurrence vector

Transformation of an e-mail text body into a word occurrence vector where each element refers to a specific word. The element is TRUE if the word occurs in the specific e-mail or FALSE if it doesn't.

Example task: spam detection (3)

hello	its	me	bye	joe	i	anna	so	why	an	class
TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	Ham
TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	Spam
FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE	Spam
TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE	Ham
FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	FALSE	Spam

Training set: A list of word occurrence vectors (instances) associated with the correct output (class labels). Is there a unique pattern for spam?

Example task: spam detection (4)

hello	its	me	bye	joe	i	anna	so	why	an	class
TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	Ham
TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	Spam
FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE	Spam
TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE	Ham
FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	FALSE	Spam

Training set: A list of word occurrence vectors (instances) associated with the correct output (class labels). Is there a unique pattern for spam?

Example task: spam detection (5)

hello	its	me	bye	joe	i	anna	so	why	an	class
TRUE	TRUE	TRUE	TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	Ham
TRUE	TRUE	FALSE	FALSE	FALSE	FALSE	TRUE	FALSE	TRUE	TRUE	Spam
FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	TRUE	FALSE	TRUE	Spam
TRUE	TRUE	FALSE	FALSE	FALSE	TRUE	FALSE	FALSE	TRUE	FALSE	Ham
FALSE	FALSE	FALSE	TRUE	TRUE	TRUE	FALSE	TRUE	FALSE	FALSE	Spam

IF me = false AND (so = true OR an = true) THEN class = spam ELSE class = ham

Tasks: the problems that can be solved in machine learning

- Supervised learning (training data provides the correct target output)
 - Spam detection is a **binary classification** task, which is the most common task in machine learning
 - In **multi-class classification**, we consider classification problems with more than two classes
 - In **regression**, we abandon the notion of discrete classes and predict a real number
- Unsupervised learning
 - In **clustering**, the task is to group data without prior information on the groups
 - **Association rules** are a kind of pattern of items that occur frequently together

Settings (problems + tasks)

- Predictive modelling (the model output involves the target)
 - Supervised learning: **classification, regression**
 - Unsupervised learning: **predictive clustering**
- Descriptive modelling
 - Supervised learning: **subgroup discovery**
 - Unsupervised learning: **descriptive clustering, association rule discovery**

Evaluating performance on a task (1)

- Many problems in computer science have a correct answer that can be verified (e.g., sorting)
- For such problems, any algorithm that is not guaranteed to produce a correct solution every time is useless
- Yet, there are often many algorithms available for solving the same problem so how do we usually tell them apart?
 - For example, by comparing their speed and memory usage
- In contrast, **most machine learning problems do not have a correct answer**
 - Read more about this and how it relates to the problem of induction and the no free lunch theorems under Background 1.1 in the main literature

Evaluating performance on a task (2)

- In machine learning, we need to get a good idea of how well an algorithm will perform on new (unseen) data or how well a model generalises to new data
- **The main performance focus is put on correctness** (of predictions, descriptions, or other solutions) rather than on traditional performance aspects, such as time and space
- This is not to say that time and space performance does not matter - **there is often a performance trade-off between correctness, space, and time** - but without sufficient correctness the other factors become unimportant
- But, how do we estimate performance on data we have not seen?

Evaluating performance on a task (3)

- Most machine learning problems are solved by training on available data with a reasonable expectation that the generated models will generalise to new data
- We often set aside some available data to use as test data
 - test data is used to simulate performance on new, unseen data
- Models that are optimised too strongly to perform well on training data may overfit
 - A clear symptom of overfitting is that the model performs well on training data but poorly on test data
- So, how do we measure correctness?

Evaluating performance on a task (4)

- There are many procedures and measures available for evaluation in machine learning
- Common evaluation measures used are accuracy (for classification) and mean squared error (for regression)
- Accuracy is the number of correctly classified instances divided by the total number of classified instances
- Mean squared error is the average of the squared differences in output between the model and the test instances
- What about clustering? How we evaluate the quality of clustering solutions?

Models: the output of machine learning (1)

- Models are what is being learned from data to solved a given task
- There are many different models to choose from
- The diversity stems from
 - the variation of the problems addressed
 - the difference in underlying principles

Models: the output of machine learning (2)

- Flach (2012) categorises basic models into three classes
 - geometric, logical, and probabilistic
- A geometric model is constructed directly in instance space, using geometric concepts such as lines, planes, and distances
 - basic linear classifier, SVM, KNN, k-means ...
- Logical models can be translated into rules that are understandable by humans
 - decision trees, ...
- Probabilistic models are based on statistical assumptions about a well-defined, but unknown, probability distribution that generates the observed data
 - Bayesian classifiers, naive Bayes, ...

Models: the output of machine learning (3)

- Flach (2012) also makes the distinction between *grouping models* and *grading models*
- The key difference between these two models is the way they handle the instance space.
- Grouping models do this by breaking up the instance space into groups or *segments*, the number of which is determined at training time.
 - tree-based models, ...
- Grading models do not employ such a notion of segment. They are able to distinguish between arbitrary instances, no matter how similar they are.
 - SVM and other geometric classifiers, ...

Features

- Features determine much of the success of a machine learning application
- Different types of features include:
 - real numbers, integers, Booleans, and categories
- Features are constructed to represent data in a convenient manner for processing, for example, when representing an e-mail message as a Boolean word occurrence vector
- Feature transformation is used to make specific aspects of data more useful (for different purposes)

References

- Flach, P. (2012). *Machine Learning: The Art and Science of Algorithms that Make Sense of Data*. Cambridge University Press.
- Japkowicz, N. & Shah, M. (2011). *Evaluating Learning Algorithms: A Classification Perspective*. Cambridge University Press.